

Introduction

1.1 Quotient Spaces

Example 1.1.1

Let $\mathbb{D}^n = \{x \in \mathbb{R}^n \mid \|x\| \leq 1\}$ be the n -disk and let $\mathbb{S}^{n-1} = \{x \in \mathbb{R}^n \mid \|x\| = 1\}$ be the $(n-1)$ -sphere. Then $\mathbb{D}^n/\mathbb{S}^{n-1}$ is homeomorphic to $\mathbb{S}^n$.

Proof. Given $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, we let $i(x) = (x_1, \dots, x_n, 0) \in \mathbb{R}^{n+1}$. We let $u = (0, \dots, 0, 1) \in \mathbb{R}^{n+1}$. Then we define:

$$f : \mathbb{D}^n \rightarrow \mathbb{S}^n, f(x) := \cos(\pi\|x\|)u + \sin(\pi\|x\|)\frac{i(x)}{\|x\|}.$$

Then f is a continuous map which induces a continuous map $\bar{f} : \mathbb{D}^n/\mathbb{S}^{n-1} \rightarrow \mathbb{S}^n$. We can then check that $\bar{f}$ is a bijection with a compact source and Hausdorff target, thus it is a homeomorphism. $\square$

Definition 1.1.2 ► Pushout

Let A, X, Y be topological spaces and $f : A \rightarrow X, g : A \rightarrow Y$ be continuous maps. We denote by $X \cup_A Y$ the **pushout** of f and g , defined as the quotient space $(X \sqcup Y)/\sim$ of the disjoint union $X \sqcup Y$ by the equivalence relation generated by $f(a) \sim g(a)$ for all $a \in A$.